

Structural Bioinf 1

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Importing PDB summary file:

```
url <- "pdb_stats.csv"
PDB_summary <- read.csv(url)
head(PDB_summary)
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	214078
2	1	0	13981
3	0	0	15759
4	3	1	4941
5	0	0	237
6	0	4	22

```
dim(PDB_summary)
```

```
[1] 6 9
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
Total <- sum(PDB_summary$Total)
X_ray <- sum(PDB_summary$X_ray)
EM <- sum(PDB_summary$EM)
```

```
100*X_ray/Total
```

```
[1] 80.95077
```

```
100*EM/Total
```

```
[1] 12.83843
```

```
#Some of the numbers are listed as strings so need to force them into numbers
#Need to remove "," (using gsub) to allow coercion to number
#Total <- sum(as.numeric(gsub(",", "", PDB_summary$Total)))
#X_ray <- sum(as.numeric(gsub(",", "", PDB_summary$X_ray)))
#EM <- sum(as.numeric(gsub(",", "", PDB_summary$EM)))
```

Percent X_ray:

```
#print(paste("Percent X-ray: ", 100*X_ray/Total))
```

Percent EM:

```
#print(paste("Percent EM: ", 100*EM/Total))
```

Q2: What proportion of structures in the PDB are protein?

```
PDB_summary$Total[1]/Total
```

```
[1] 0.8596889
```

```
#Total_Protein <- sum(as.numeric(gsub(",", "", PDB_summary$Total[1:3])))
#print(paste("Percent Protein Structures: ", 100 * Total_Protein/Total))
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

4,934 Structures

HIV-1 protease

1HSG

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Protons (H) are not shown

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

HOH 308?

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

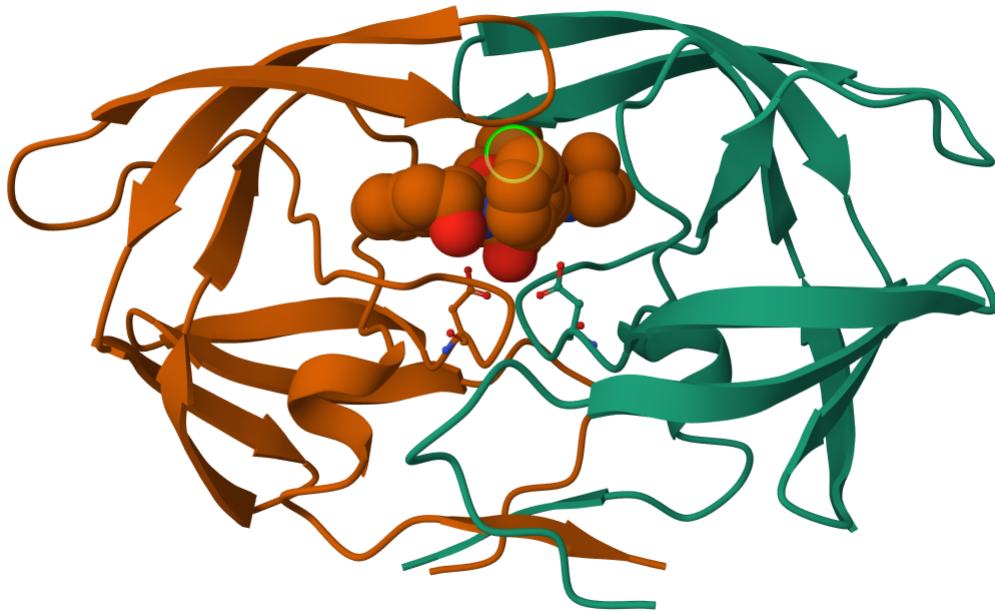


Figure 1: HIV protease

Discussion Topic: Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site?

Presumably via protein dynamics

Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

beta-strands T96-N98 form a sheet only in the dimer (Turned on colors by chain)

3. Introduction to Bio3D in R

```
library(bio3d)
```

Reading PDB file data into R

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
  Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172  (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198 (Ca atoms)

Q8: Name one of the two non-protein residues?

HOH (127), MK1 (1)

Q9: How many protein chains are in this structure?

Chains#: 2

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

```
head(pdb$helix)
```

```
$start
```

```
87 87
```

```
$end
```

```
90 90
```

```
$chain
```

```
[1] "A" "B"
```

```
$type
```

```
[1] "1" "1"
```

Quick PDB visualization in R

```
# install.packages("pak")  
#pak::pak("bioboot/bio3dview")
```

Not able to install bioboot/bio3dview in the way shown. Needed to run the following in the console: `install.packages("devtools")` `devtools::install_bitbucket("Grantlab/bio3d-view")` (note: the resulting package is named "bio3d.view") Possibly: `remotes::install_bitbucket("Grantlab/bio3d-view")` [may need `install.packages("rlang")`] `install.packages("rgl")`

```
library(bio3d.view)  
library(NGLViewerR)  
  
view.pdb(pdb) |>  
  setSpin()
```

Computing connectivity from coordinates...

```
$x  
$x$setSpin  
[1] TRUE
```

This opens up in XQuartz

```
# Select the important ASP 25 residue  
sele <- atom.select(pdb, resno=25)  
  
# and highlight them in spacefill representation  
view.pdb(pdb, cols=c("navy","teal"),  
  highlight = sele,  
  highlight.style = "spacefill") |>  
  setRock()
```

Computing connectivity from coordinates...

```
$x
$x$setRock
[1] TRUE
```

Didn't see a difference in the XQuartz display

Predicting functional motions of a single structure

Let's read a new PDB structure of Adenylate Kinase and perform Normal mode analysis.

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

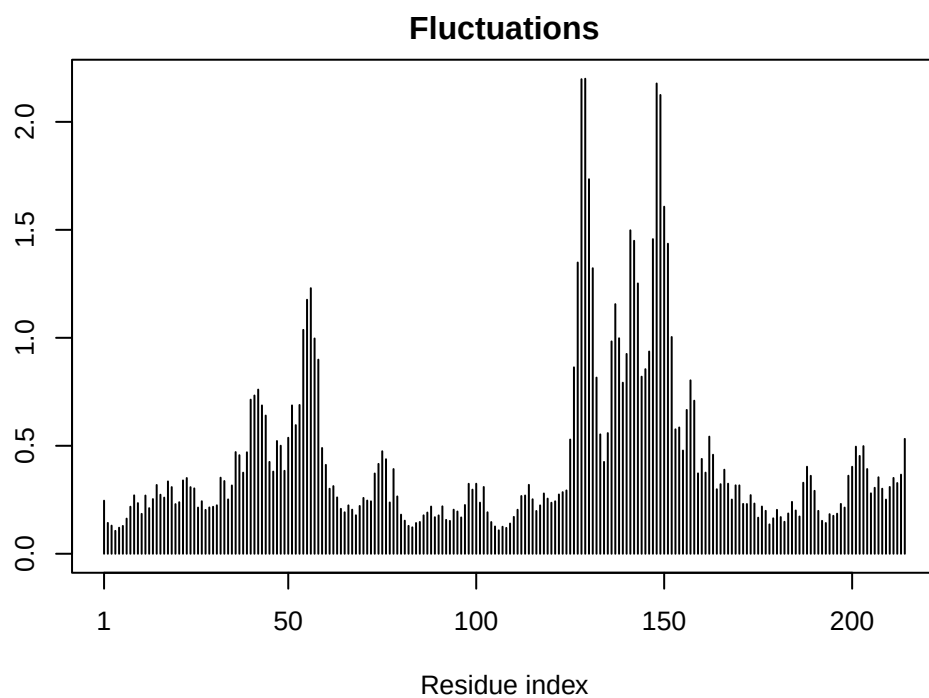
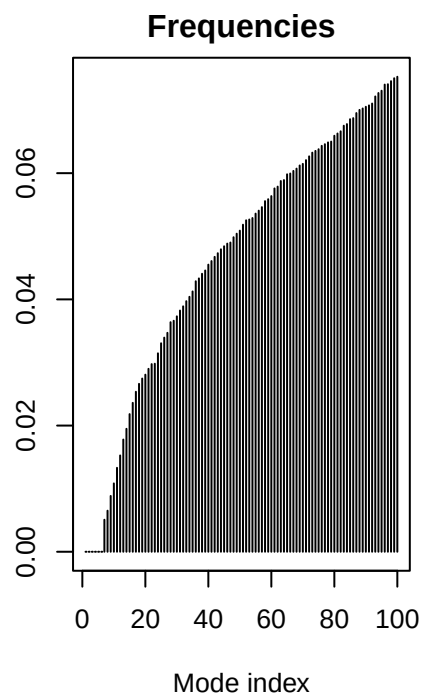
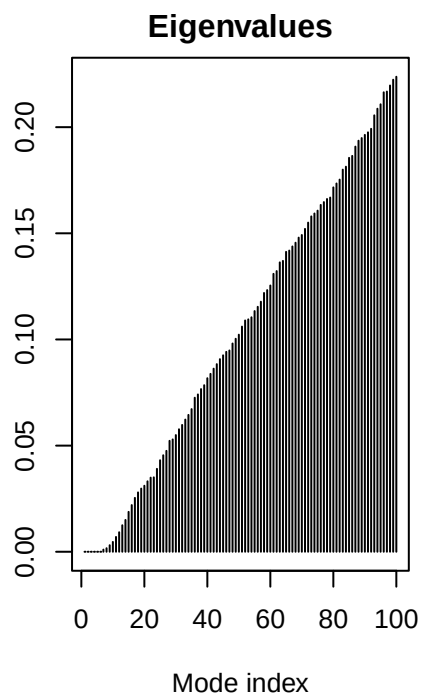
```
MRILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
# Perform flexibility prediction  
m <- nma(adk)
```

```
Building Hessian...      Done in 0.013 seconds.  
Diagonalizing Hessian... Done in 0.062 seconds.
```

```
plot(m)
```

Note: gave initial “figure margins too large” error; fixed by enlarging the source window

```
mktrj(m, file="adk_m7.pdb")
```

Tried to open in Mol* but received ‘Error opening file ‘adk_m7.pdb’ Was able to open and play it in PyMol

```
library(bio3d.view)
#view.nma(m, pdb=adk)
```

Couldn’t make view.nma work

4. Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

“msa”

Q11. Which of the above packages is not found on BioConductor or CRAN?

“bioboot/bio3dview”

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

Search and retrieve ADK structures

```
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60
      61      .      .      .      .      .      120
```

```

pdb|1AKE|A    DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDRI
              61          .          .          .          .          .          120

              121          .          .          .          .          .          180
pdb|1AKE|A    VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM TAPLIG
              121          .          .          .          .          .          180

              181          .          .          .          214
pdb|1AKE|A    YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
              181          .          .          .          214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 aa

```
# Blast or hmmer search
#b <- blast.pdb(aa)
```

```
# Plot a summary of search results
#hits <- plot(b)
#same as plot.blast(b)
#fig-height = 7
```

```
hits <- NULL
```

```
hits$ pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6H
```

```
#plot.blast(b)
```

```
# List out some 'top hits'
```

```
head(hits$ pdb.id)
```

```
[1] "1AKE_A" "6S36_A" "6RZE_A" "3HPR_A" "1E4V_A" "5EJE_A"
```

```
length(hits$pdb.id)
```

```
[1] 13
```

```
# Download releated PDB files  
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/5EJE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4Y.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3X2S.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6HAP.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6HAM.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/4K46.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3GMT.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/4PZL.pdb.gz exists. Skipping download
```

		0%
=====		8%
=====		15%
=====		23%
=====		31%
=====		38%
=====		46%
=====		54%
=====		62%
=====		69%
=====		77%
=====		85%
=====		92%
=====		100%

```
# Align related PDBs  
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

pdb/split_chain/1AKE_A.pdb
pdb/split_chain/6S36_A.pdb
pdb/split_chain/6RZE_A.pdb
pdb/split_chain/3HPR_A.pdb
pdb/split_chain/1E4V_A.pdb
pdb/split_chain/5EJE_A.pdb
pdb/split_chain/1E4Y_A.pdb
pdb/split_chain/3X2S_A.pdb
pdb/split_chain/6HAP_A.pdb
pdb/split_chain/6HAM_A.pdb
pdb/split_chain/4K46_A.pdb
pdb/split_chain/3GMT_A.pdb
pdb/split_chain/4PZL_A.pdb

 PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
.. PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
...

Extracting sequences

pdb/seq: 1 name: pdb/split_chain/1AKE_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2 name: pdb/split_chain/6S36_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3 name: pdb/split_chain/6RZE_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4 name: pdb/split_chain/3HPR_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5 name: pdb/split_chain/1E4V_A.pdb
pdb/seq: 6 name: pdb/split_chain/5EJE_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7 name: pdb/split_chain/1E4Y_A.pdb
pdb/seq: 8 name: pdb/split_chain/3X2S_A.pdb
pdb/seq: 9 name: pdb/split_chain/6HAP_A.pdb
pdb/seq: 10 name: pdb/split_chain/6HAM_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11 name: pdb/split_chain/4K46_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE

```
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb
```

—This could not be rendered

```
library(bio3d.view)

view.pdbs(pdbs)
```

```
# Vector containing PDB database codes
ids <- basename.pdb(pdbs$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-ray

6S36_A ray	6S36	A	Protein	214	X-
6RZE_A ray	6RZE	A	Protein	214	X-
3HPR_A ray	3HPR	A	Protein	214	X-
1E4V_A ray	1E4V	A	Protein	214	X-
5EJE_A ray	5EJE	A	Protein	214	X-
1E4Y_A ray	1E4Y	A	Protein	214	X-
3X2S_A ray	3X2S	A	Protein	214	X-
6HAP_A ray	6HAP	A	Protein	214	X-
6HAM_A ray	6HAM	A	Protein	214	X-
4K46_A ray	4K46	A	Protein	214	X-
3GMT_A ray	3GMT	A	Protein	230	X-
4PZL_A ray	4PZL	A	Protein	242	X-

	resolution	scopDomain	pfam
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
6S36_A	MG (2),NA,CL (3)
6RZE_A	CL (2),NA (3)
3HPR_A	AP5

1E4V_A	AP5
5EJE_A	CO,AP5
1E4Y_A	AP5
3X2S_A	AP5,MG,JPY (2)
6HAP_A	AP5
6HAM_A	AP5
4K46_A	PO4,AMP,ADP
3GMT_A	SO4 (2)
4PZL_A	GOL,FMT,CA

	ligandName
1AKE_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6S36_A	MAGNESIUM ION (2),SODIUM ION,CHLORIDE ION (3)
6RZE_A	CHLORIDE ION (2),SODIUM ION (3)
3HPR_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
1E4V_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
5EJE_A	COBALT (II) ION,BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
1E4Y_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
3X2S_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION,N-(pyren-1-ylmethyl)acetamide (2)
6HAP_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6HAM_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
4K46_A	PHOSPHATE ION,ADENOSINE MONOPHOSPHATE,ADENOSINE-5'-
DIPHOSPHATE	
3GMT_A	SULFATE ION (2)
4PZL_A	GLYCEROL,FORMIC ACID,CALCIUM ION

	source
1AKE_A	Escherichia coli
6S36_A	Escherichia coli
6RZE_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
4K46_A	Photobacterium profundum

3GMT_A Burkholderia pseudomallei 1710b
 4PZL_A Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB.
 6S36_A
 6RZE_A
 3HPR_A
 1E4V_A
 loop
 5EJE_A
 1E4Y_A
 loop
 3X2S_A
 conjugated adenylate kinase
 6HAP_A
 6HAM_A
 4K46_A
 3GMT_A
 4PZL_A

Cryst

The crys

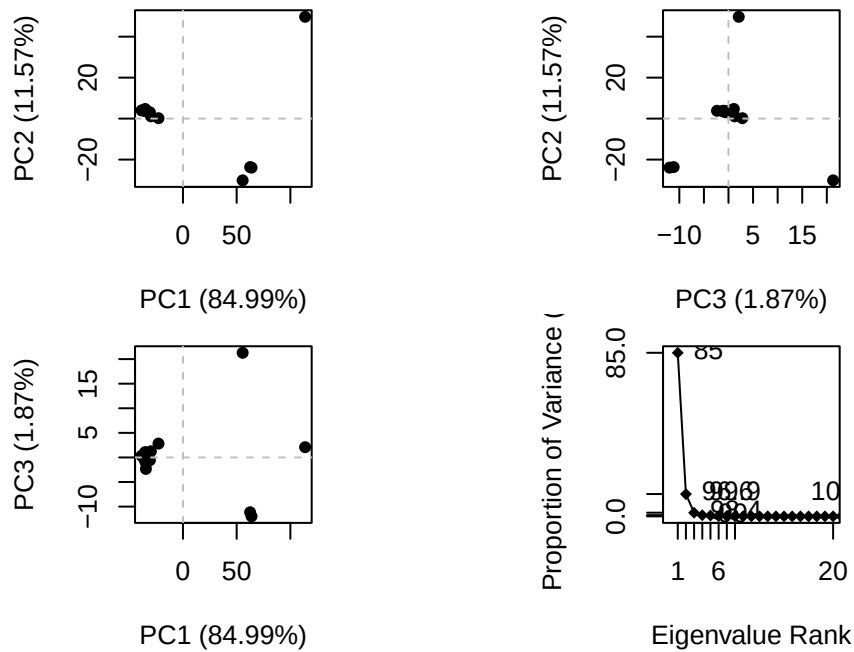
		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al.	J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al.	Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al.	Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al.	Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al.	Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al.	Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al.	Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al.	Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al.	To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al.	Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al.	To be published	0.19360	0.23680

	rWork	spaceGroup
1AKE_A	0.19600	P 21 2 21
6S36_A	0.15940	C 1 2 1
6RZE_A	0.18190	C 1 2 1
3HPR_A	0.20620	P 21 21 2
1E4V_A	0.19600	P 21 2 21
5EJE_A	0.18630	P 21 2 21
1E4Y_A	0.17800	P 1 21 1
3X2S_A	0.20700	P 21 21 21
6HAP_A	0.22370	I 2 2 2

```
6HAM_A 0.20311      P 43
4K46_A 0.16730 P 21 21 21
3GMT_A 0.23500      P 1 21 1
4PZL_A 0.19130      P 32
```

Principal component analysis

```
# Perform PCA
pc.xray <- pca(pdbbs)
plot(pc.xray)
```

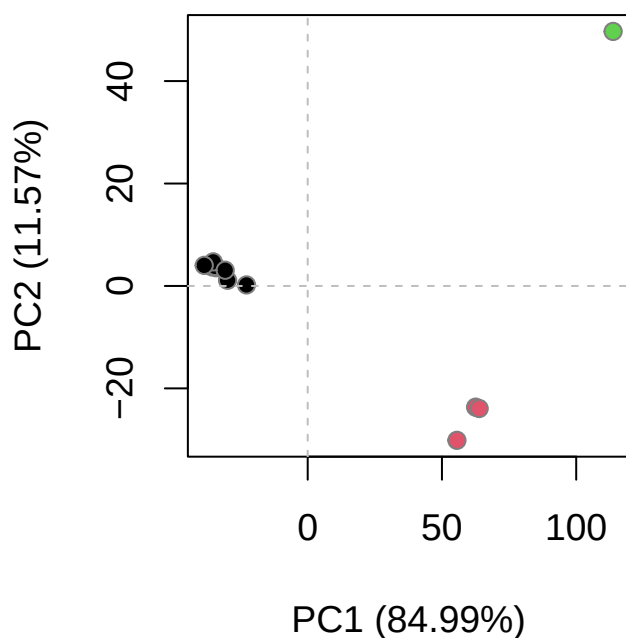


```
# Calculate RMSD
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



5. Optional further visualization

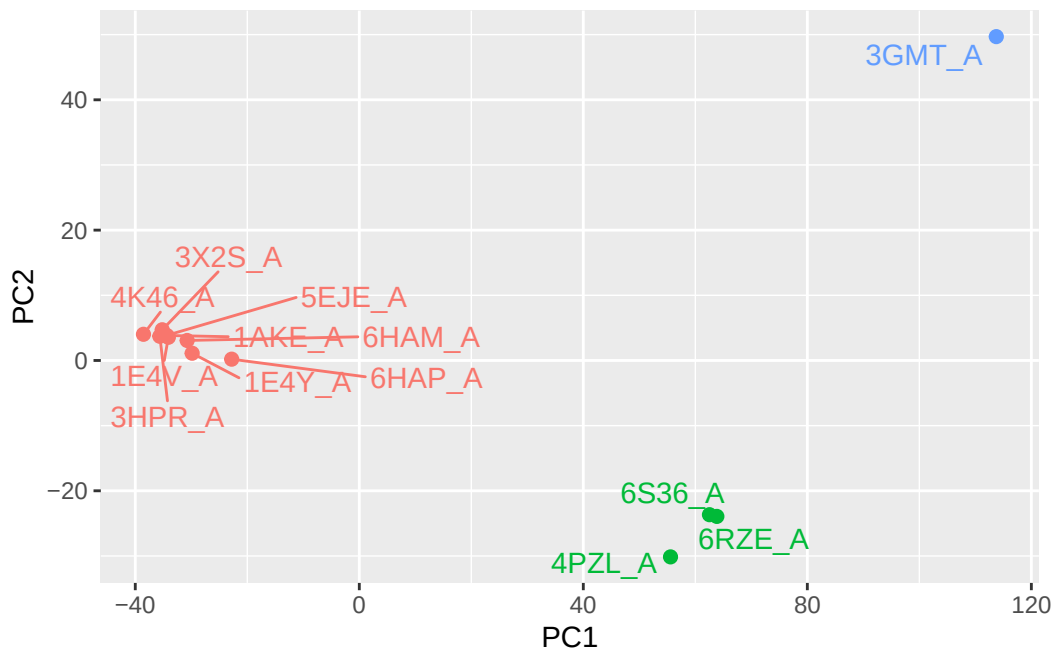
```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

again, could open the file with PyMol but not Mol

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                  PC2=pc.xray$z[,2],
                  col=as.factor(grps.rd),
                  ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



6. Normal mode analysis [optional]

```
# NMA of all structures
modes <- nma(pdbbs)
```

Details of Scheduled Calculation:

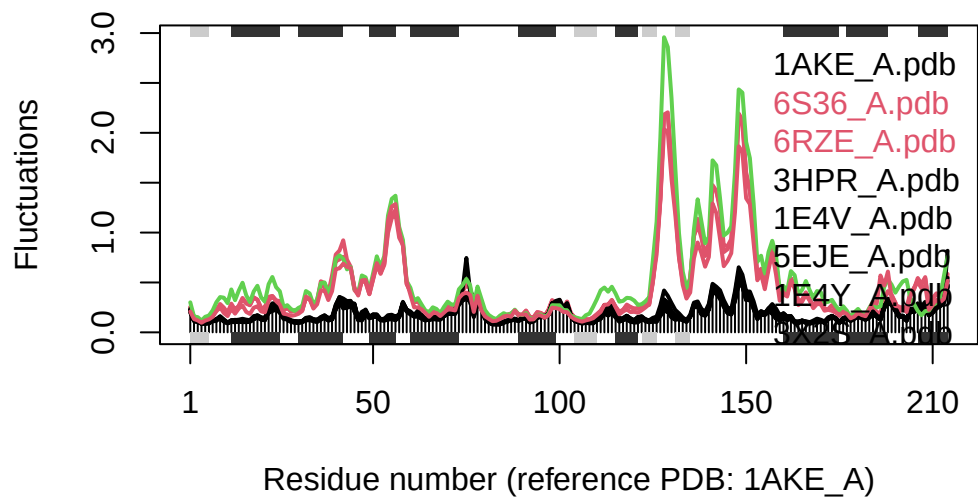
```
... 13 input structures
... storing 606 eigenvectors for each structure
... dimension of x$U.subspace: ( 612x606x13 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 36.9 Mb
```

		0%
=====		8%
=====		15%
=====		23%



```
plot(modes, pdbc, col=grps.rd)
```

Extracting SSE from pdbc\$sse attribute



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

proteins colored by PCA groups (grps.rd) differ most around aa 60 and aa 135